Gradient Descent Finds the Cubic-Regularized Nonconvex Newton Step

Yair Carmon and John C. Duchi

SIAM Journal on Optimization, 2019 (preprint 2016)

List item 46 / Ref 71 | Tags: G, D, E / F, B | Tier: Skim (targeted deep reading)

The paper in one minute

Question: Can plain gradient descent solve the nonconvex cubic-regularized quadratic subproblem used inside Newton-type methods?
Model/Method: Study a possibly indefinite quadratic plus a positive cubic norm term; analyze GD in the eigenbasis of its Hessian.
Finding: Under specified initialization and step-size conditions, GD approaches the *global* subproblem optimum at a quantified rate; a small random perturbation handles a hard alignment case.
Why we care: A second-order saddle-escaping Newton step can itself be approximated with simple matrix-vector/gradient computations.

1. Start with the task Newton's method leaves unsolved

A standard Newton step minimizes a quadratic Taylor approximation. When the Hessian has negative eigenvalues, that quadratic is unbounded below: move far enough along a negative-curvature direction and its value keeps decreasing. Cubic regularization adds a stabilizing term to make the subproblem bounded below:

$$q(s) = b^\top s + \frac{1}{2} s^\top A s + \frac{\rho}{3}\|s\|^3, \qquad A = A^\top,\ \rho > 0.$$

Here b acts like the gradient of the original objective, A its Hessian and s the proposed update. Although the cubic term keeps q from decreasing without bound, q is generally *still nonconvex*. The question is why the cheapest possible solver—GD—can nevertheless find its global optimum.

Worked example: a scalar negative-curvature model

Choose $A = -1$, $b = 0$, $\rho = 1$, with a single coordinate s:

$$q(s) = -\tfrac{1}{2}s^2 + \tfrac{1}{3}|s|^3, \quad q'(s) = -s + s|s|.$$

Stationary points are $s = 0, \pm 1$. The origin is a local *maximum* ($q''(0) = -1$), whereas $q(\pm 1) = -1/6 < q(0) = 0$. Run GD with $\eta = 0.25$ from $s_0 = 0.2$:

$$s_1 = 0.2 - 0.25(-0.2 + 0.04) = 0.24, \qquad s_2 = 0.24 - 0.25(-0.24 + 0.0576) = 0.2856.$$

The iterate moves away from the negative-curvature stationary point and toward $s = +1$. Starting *exactly* at zero would leave GD stuck; this elementary case foreshadows why initialization matters.

Important precision. This is a pedagogical one-dimensional example, not proof of the full theorem. The general problem may have many dimensions, unequal eigenvalues, saddle points and a bad local minimum.

2. Why the eigenvectors explain gradient-descent dynamics

Differentiate the actual cubic objective. Since $\nabla(\|s\|^3/3) = \|s\|s$, the full update is

$$\nabla q(s) = As + b + \rho\|s\|s, \qquad s_{t+1} = s_t - \eta(As_t + b + \rho\|s_t\|s_t).$$

Diagonalize $A = V\Lambda V^\top$. In eigen-coordinate i, this gives the recurrence

$$s_{t+1}^{(i)} = \big(1 - \eta\lambda_i(A) - \eta\rho\|s_t\|\big)s_t^{(i)} - \eta b^{(i)}.$$

A negative λ_i initially *amplifies* the corresponding component. Once $\|s_t\|$ grows enough, the cubic term changes the multiplier so the trajectory can contract toward the global minimizer. The paper's proof separates these two stages: growth in the most negative eigendirection, then convergence in the regularized region.

What Theorems 3.1 and 3.2 establish

Theorem 3.1 (deterministic). For a sufficiently small fixed step, the prescribed initialization aligned with $-b$, and nonzero projection of b onto a smallest-eigenvalue direction, GD reaches objective error $q(s_t) - q(s_*) \le \varepsilon$ after an explicit number of steps. The analysis exhibits different accuracy/conditioning regimes: roughly $\widetilde{O}(1/\varepsilon)$ in a difficult regime, an intermediate $1/\sqrt{\varepsilon}$ behavior, and logarithmic dependence on $1/\varepsilon$ in the favorable local regime. **Theorem 3.2 (randomized hard case).** If b is orthogonal to the most-negative eigenspace, the deterministic alignment argument can fail. Slightly perturbing b before running GD makes the needed projection nonzero with high probability, at a cost involving logarithms of the dimension and failure probability.

A useful contrast with ordinary Newton's method

Ordinary Newton solves $As = -b$, which is not a minimization guarantee when A is indefinite. Cubic regularization instead minimizes a *stabilized* nonlinear model. Crucially, this paper studies a solver for *that inner model*, not convergence of unconstrained GD on every original nonconvex objective.

What to read: Original Sections 2–3 (assumptions, Proposition 2.5, Theorems 3.1–3.2); Section 6 for the outer-method application; Section 5 only for line-search nuances. Proof Sections 4 and appendix can be skimmed for the two-stage argument.

3. Contribution, qualifications, and survey fit

What is new

- A finite-iteration analysis of plain GD on an *indefinite quadratic plus cubic regularizer*, yielding a surprisingly strong global subproblem guarantee.
- A treatment of the "hard case," in which the linear term has no component in a most-negative eigendirection, via a randomized perturbation.
- An outer cubic-Newton application using Hessian-vector products rather than forming or inverting a dense Hessian.

Assumptions and limitations

The primary analysis uses symmetric A, $\rho > 0$, carefully limited constant step size (Assumption A), and special initialization $s_0 = -rb/\|b\|$ (Assumption B); randomized results modify the hard case. The outer optimization application requires regularity such as a Lipschitz Hessian. None of this proves that arbitrary initialization, arbitrary learning rates or ordinary neural-network GD always finds global minima. The subproblem can have spurious local minima even though *its special GD trajectory* reaches a global solution.

Cross-paper synthesis

Dauphin (2014) also uses curvature to motivate saddle-aware optimization; Carmon–Duchi focuses on efficiently solving a stabilizing cubic model. **Wang et al. (2020), Ref 72** adds momentum to the *outer* cubic-regularization method, not to precisely this convergence proof. **Jin (2017)** uses gradient perturbations directly on the objective instead of solving cubic subproblems.

Quick reference

Object	Meaning
$\frac{1}{2}s^TAs + b^Ts + \frac{\rho}{3}\|s\|^3$	Cubic-regularized Newton model: second-order Taylor terms plus stabilizer.
$As + b + \rho\|s\|s$	Gradient used in inexpensive inner GD steps.
$\lambda_{\min}(A) < 0$	Negative curvature makes the unregularized quadratic nonconvex.
ε-accurate step	Objective gap to *global subproblem optimum*, not merely small gradient.
Hard case	Linear term orthogonal to the least-eigenvalue eigenspace; perturbation resolves alignment.

Papers Master note

Carmon–Duchi show that GD, under controlled initialization/step sizes, globally solves cubic-regularized indefinite quadratic subproblems to stated accuracy. Random perturbation treats the hard alignment case. This provides a matrix-free inner solver for second-order saddle-escaping methods, not a global neural-network GD theorem.

Source: original full text, Sections 2–3 and 6; SIAM J. Optim. 29(3), 2019. Survey tag: G, D, E / F, B.